

2 The database

CARINA consists of a collection of 185 hydrographic cruises or projects, divided into 3 regions, the Atlantic (CARINA-ATL), the Arctic Mediterranean Seas (CARINA-AMS) and the Southern Ocean (CARINA-SO) regions. This paper focuses on the oxygen data from the Atlantic Ocean, which is the region with the highest number of cruises (98 cruises). The CARINA-ATL region has four cruises in common with the CARINA-AMS group and four cruises in common with the CARINA-SO group, in order to ensure consistency also across the three regions.

For the secondary quality control, 6 hydrographic cruises from the World Ocean Circulation Experiment (WOCE) as synthesized by GLODAP (Key et al., 2004) were added to ensure consistency with this prior synthesis effort. However, the data from these reference cruises are not included in the final product. 29 cruises from the CARINA-ATL database were identified as core cruises, primarily on the basis of their having a good spatial distribution and a higher expected data quality. During the computation of the adjustments, more weight was given to these core cruises as well as to the 6 WOCE cruises, ensuring that cruises with lower data quality are not causing unwarranted adjustments in the cruises with higher data quality.

2.1 Oxygen data

The CARINA-ATL database contains a total of 113005 oxygen samples from 9535 stations. A map of the locations of the hydrographic stations with at least one oxygen sample (Fig. 1) shows that CARINA-ATL has a good spatial distribution, with the densest coverage in the northeast Atlantic. The data span three decades, with the earliest samples stemming from 1977 and the latest samples taken in 2006 (Fig. 2). The majority of the data stem from the 1990s, with a reasonably good coverage between about 1990 and 2005. The years 1997, 1998, 2001 and 2003 have the greatest number of samples, largely driven by the fact that these years include basin-wide transects (Fig. 3). Maps of the data from 1977 to 2007 with a time step of 5 years show that from 1992 to 2007

107

the spatial coverage is good, and particularly in the last five years is dominated by long transects. From 1987 to 1991 the data are especially concentrated in the northeastern part, while from 1977 to 1986 there are only few data stemming from along the Iberian coast and in the Arctic. The only cruise in 1977 (29CS19771007) is one of the Galicia cruises (Galicia4) and has 88 stations with oxygen data only in the upper 1500 m. For this reason, we did not include this cruise in the secondary QC, i.e. no adjustment is recommended (http://cdiac.ornl.gov/oceans/CARINA/Carina_inv.html and table cruise summary from Key et al. (2009)).

In order to provide an overview of the oxygen distribution, we plot in Fig. 4 the oxygen data versus depth for 11 large-scale regions, separating the time periods by color. This plot reveals, for example, that all oxygen data before 1986 are from the upper 2000 m only (green dots in the panel d). The vertical distribution of oxygen in the Atlantic Ocean reflects the combined effects of gas exchange, of the photosynthesis/remineralization cycle, and of ocean circulation. Near the surface ocean, rapid gas exchange keeps oxygen very near the saturation concentration (Sarmiento and Gruber, 2006), despite oxygen production from photosynthesis and oxygen demand by the upwelling/mixing of low oxygen waters from below. In the interior, the oxygen demand from the remineralization of the sinking organic matter leads to decreasing oxygen concentrations, but this decrease is modified strongly by the level by which ocean circulation and mixing are able to transport oxygen from the upper ocean into the deep. For example, oxygen rich water penetrates into the deep North Atlantic as a result of the southward spreading of recently ventilated North Atlantic Deep Water. This easily identifiable water mass that occupies the depth between 1000 m to 4000 m and extends southward from the Labrador Sea to the Antarctic Divergence (Fig. 4, panel c, e, and g). The Labrador Sea Water (LSW) is also well distinguishable in the Labrador Sea as a huge volume of nearly homogeneous water with high oxygen content. This water mass is a product of winter deep convection (Fig. 4 northwest region, panel a). In the intermediate-depth eastern tropical ocean (Fig. 4 the eastern region in the tropical zones, panel f) it is also possible to identify the horizontal oxygen minimum zone (OMZ).